

S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

– Hira Kumar

QBASIC

❖ Selection(decision) statement

- In qbasic is used to make decision. It allows the program to execute certain blocks of code depending on whether a condition is True or False.
- If...else, if, if...else...if,... is a conditional statement used to make decisions.
- It executes code only once depending on the condition.
- It does not repeat, so it is not a loop.

1. IF

i. IF.....Else.....End if

Syntax :-

If Condition Then

-----Statements (true)

Else

-----Statements (false)

End If

Q. WAP to check eligible for Vote or Not.

```
1 Cls
2 Print "Goverment Degree College, Rajgir"
3 Print "======"
4 Input " Enter any number"; a
5 If a >= 18 Then
6     Print "you are eligible"
7 Else
8     Print "you are not eligible"
9 End If
10 End
11
```

Untitled

Goverment Degree College, Rajgir
=====
Enter any number? 20
you are eligible

ii. IF Without ELSE

Syntax :-

IF Condition then

End If

Q. Write a program to check odd or even ?

```
1 Cls
2 Print "Goverment Degree College, Rajgir"
3 Print "======"
4 Input " Enter any number"; a
5 If a Mod 2 = 0 Then
6     Print a" is even number"
7 End If
8 End
9
```

Untitled

Goverment Degree College, Rajgir
=====
Enter any number? 10
10 is even number

iii. Nexted IF.....ELSE

Syntax :-

If Condition then

-----statement

Else

If Condition

-----statement

Else

-----statement

End if

End if

q. WAP to take input and check this number +ve, -ve or 0.



QB64 x64

File Edit View Search Run Debug Options Tools

Untitled*

```
1 Cls
2 Print "Goverment Degree College, Rajgir"
3 Print "=====
4 Input " Enter any number"; a
5 If a > 0 Then
6     Print a; "is +ve number"
7 Else
8     If a < 0 Then
9         Print a; "is -ve number"
10    Else
11        Print "this is 0"
12    End If
13 End If
14 End
15
```

Untitled

Goverment Degree College, Rajgir

=====

Enter any number? -20

-20 is -ve number

❖ Loops In QBASIC

- A program that executes a block of code many times until the given condition becomes false.

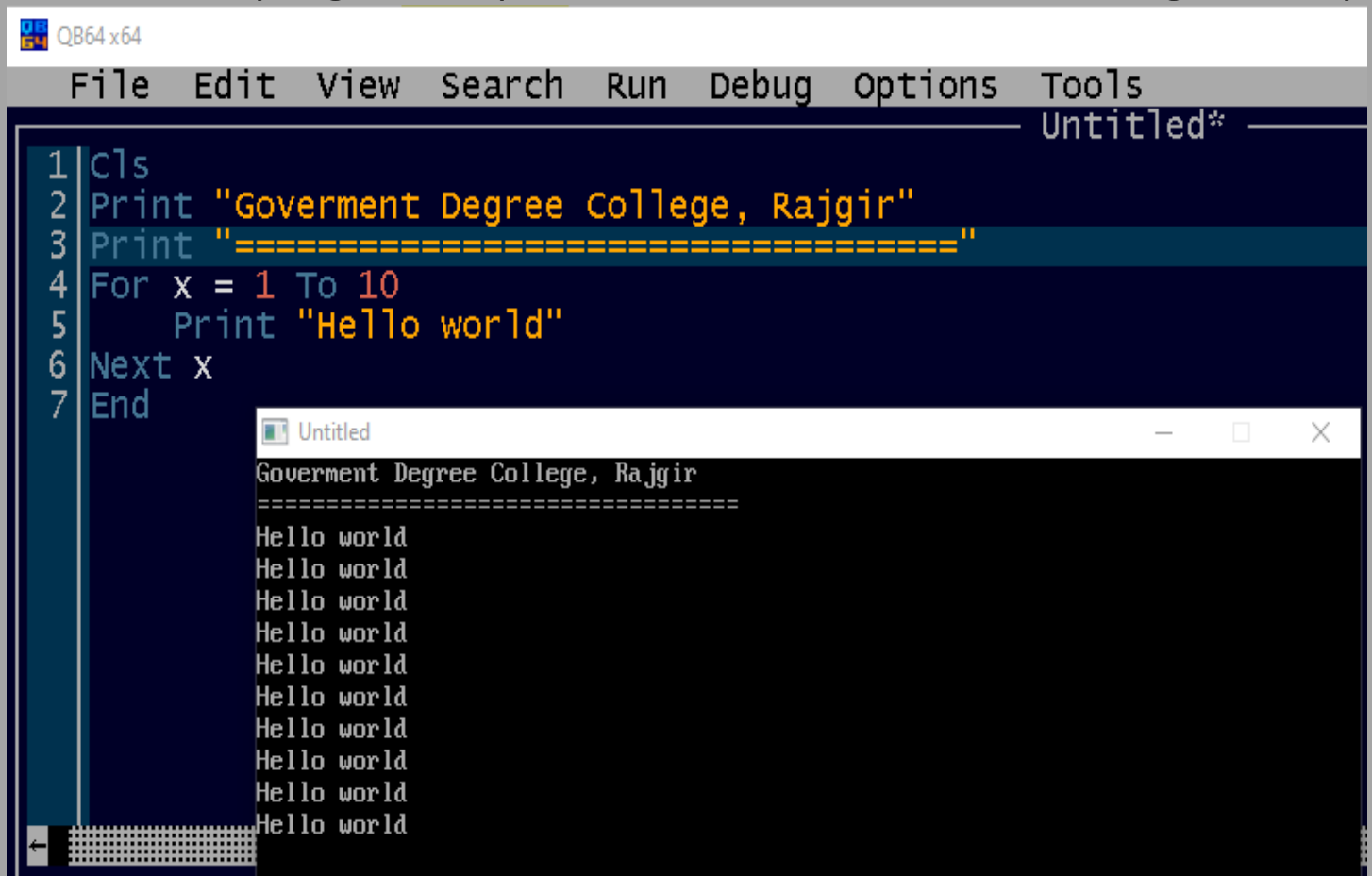
2. FOR.....NEXT Loop

It is used to execute a block of code repeatedly for a specific number of times.

Syntax :-

For variable=Condition [Step Increment] (default step 1)
----- (statement)
Next Variable

Q1. Write a program to print "hello world" 10 times using For loop.

The image shows a screenshot of the QBASIC 64-bit environment. The main window, titled 'Untitled*', contains a program with the following code:

```
1 Cls
2 Print "Goverment Degree College, Rajgir"
3 Print "=====
4 For x = 1 To 10
5     Print "Hello world"
6 Next x
7 End
```

The program is executed, and a separate output window titled 'Untitled' displays the results. The output consists of the text 'Goverment Degree College, Rajgir' followed by a line of 10 equals signs, and then the text 'Hello world' printed 10 times, one on each line.

Q WAP to print star this format.

```

1 Cls
2 For i = 1 To 5
3     For j = 1 To i
4         Print "*";
5     Next j
6     Print
7 Next i
8 End

```

Output window shows:

```

*
**
***
****
*****

```

3. DO ____loop

- Do.....loop in qbasic is used for repeating a block of code indefinitely or until a certain condition is met.

Do while....Loop (OR) Do.....Loop while:- Executes the loop as long as the condition is true.

Syntax :-

```

DO While condition
-----
Loop
(1st check condition after statement)

```

Syntax :-

```

Do
-----
Loop while Condition
(1st check statement after condition)

```

```

1 Cls
2 Print "Government Degree College, Rajgir"
3 Print "=====
4 x = 1
5 Do While x <= 10
6     Print x;
7     x = x + 1
8 Loop
9
10 End

```

Output window shows:

```

Government Degree College, Rajgir
=====
1 2 3 4 5 6 7 8 9 10

```

```

1 Cls
2 Print "Government Degree College, Rajgir"
3 Print "=====
4 x = 1
5 Do
6     Print x
7     x = x + 1
8 Loop while x <= 10
9 End

```

Output window shows:

```

Government Degree College, Rajgir
=====
1
2
3
4
5
6
7
8
9
10

```

DO.....Loop Until

- Executes the loop until the condition becomes true. OR
- Executes the loop continues as long as the condition is false.

Syntax

Do

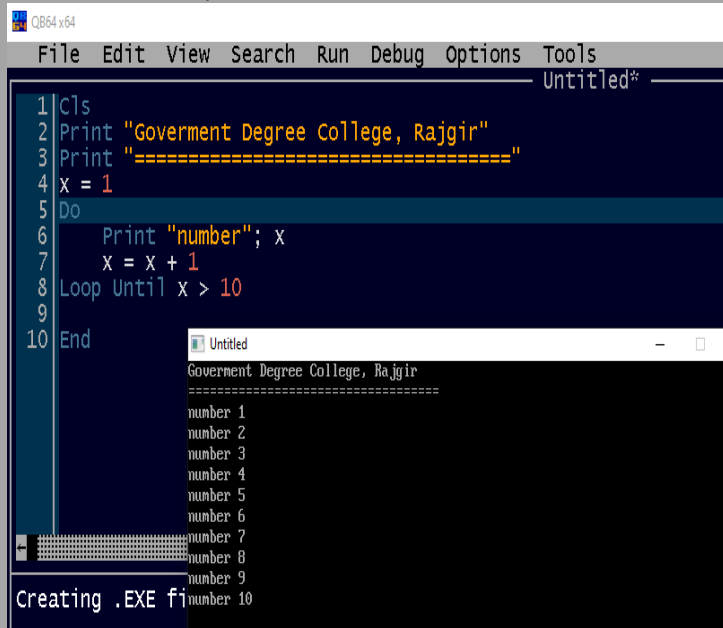
Loop until Condition

(1st check statement after condition)

Do until Condition

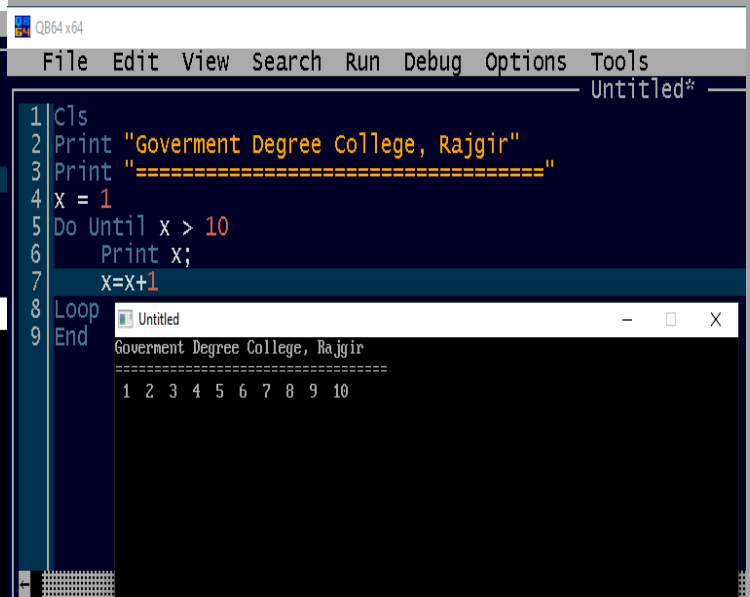
Loop

(1st check condition after statement)



```
1 Cls
2 Print "Government Degree College, Rajgir"
3 Print "=====
4 x = 1
5 Do
6   Print "number"; x
7   x = x + 1
8 Loop Until x > 10
9
10 End
```

Government Degree College, Rajgir
=====
number 1
number 2
number 3
number 4
number 5
number 6
number 7
number 8
number 9
number 10



```
1 Cls
2 Print "Government Degree College, Rajgir"
3 Print "=====
4 x = 1
5 Do Until x > 10
6   Print x;
7   x=x+1
8 Loop
9 End
```

Government Degree College, Rajgir
=====
1 2 3 4 5 6 7 8 9 10

Infinite loop with Exit do

An infinite loop can be controlled using EXIT DO.

Syntax –

DO

----- (statement)

If ----- (Condition) then Exit Do

loop

The screenshot shows the QB64 x64 IDE with a menu bar (File, Edit, View, Search, Run, Debug, Options, Tools) and a toolbar. The main window, titled 'Untitled*', contains the following BASIC code:

```
1 Cls
2 Print "Government Degree College, Rajgir"
3 Print "=====
4 Print " For Exit Press 1"
5 Do
6     Input "number"; a
7     If a = 1 Then Exit Do
8     Print "You entered", a
9 Loop
10
11 End
```

A secondary window titled 'Untitled' displays the program's output:

```
Government Degree College, Rajgir
=====
For Exit Press 1
number? 2
You entered    2
number? 10
You entered    10
number? 5
You entered    5
number? 1
```

Q3. WAP to print reverse number 20 to 1 using do____loop while.

Q4. WAP to print table of 10 using do loop.

4. While____Wend

- While.....Wend loop, if the condition is true, all statements are executed until WEND. If the Condition is false, the loop is terminated and the control jumps to next statement after WEND.

Syntax

```
While Condition
    -----(statement)
Wend
```

Q1. WAP to print 1 to 10 using While loop.



```
1 Cls
2 Print "Goverment Degree College, Rajgir"
3 Print "======"
4 X = 1
5 While X <= 10
6     Print X
7     X=X+1
8 Wend
9 End
```

Untitled

Goverment Degree College, Rajgir

=====

```
1
2
3
4
5
6
7
8
9
10
```

5. GOTO Statement

- Jump Statement – Can make the control of the program jump from one part to another.
- Uses label to identify different parts e.g – A: (it is lable,lable must end colon (:)) symbol)
- Can be used to make loops.

Q1. Write a program use GOTO statement


```
QB64 x64
File Edit View Search Run Debug Options Tools
Untitled*

1 Cls
2 Print "Government Degree College, Rajgir"
3 Print "=====
4 Print "Print Line 1"
5 GoTo A:
6 Print "Print Line 2"
7 Print "Print Line 3"
8 a: Print "Print Line 4"
9 End
```

```
Untitled
Government Degree College, Rajgir
=====
Print Line 1
Print Line 4
```

Q2. WAP to print 1 to 20 using GOTO

```
QB64 x64
File Edit View Search Run Debug Options Tools
Untitled*

1 Cls
2 Print "Government Degree College, Rajgir"
3 Print "=====
4 x = 1
5 a:
6 Print x
7 x = x + 1
8 If x <= 10 GoTo a:
9 End
```

```
Untitled
Government Degree College, Rajgir
=====
1
2
3
4
5
6
7
8
9
10
```

Creating .EXE

6. SELECT.....CASE Statement

This statement is used to handle multiple conditions.

Syntax

Select condition variable

Case 1

Case 2

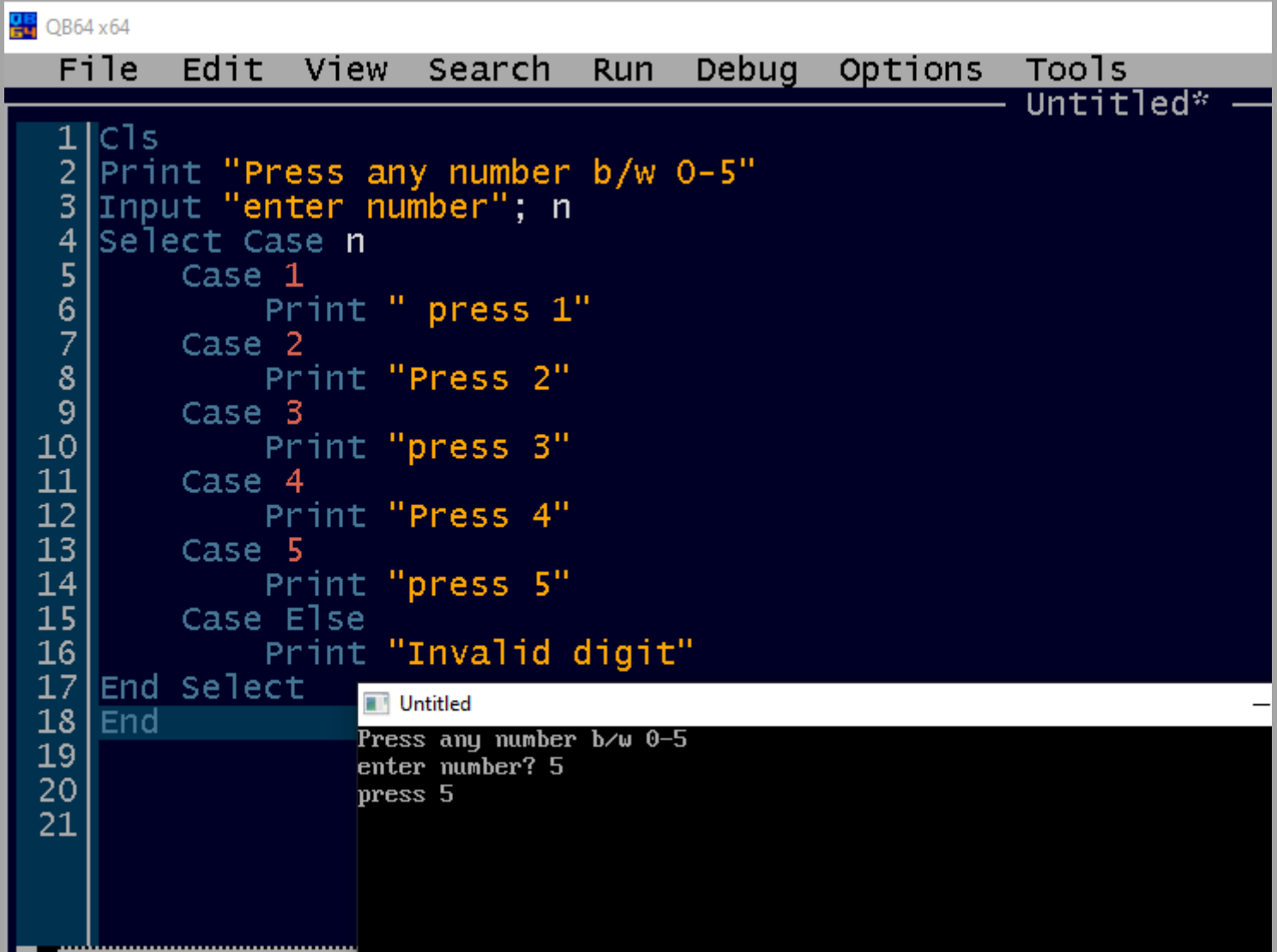
Case 3

Case 4

Case Else

End Select

Q. WAP to take input digit by user 0 to 5 (0-5) other input is invalid.



The screenshot shows the QB64 x64 IDE with a BASIC program for digit validation. The code is as follows:

```
1 Cls
2 Print "Press any number b/w 0-5"
3 Input "enter number"; n
4 Select Case n
5     Case 1
6         Print " press 1"
7     Case 2
8         Print "Press 2"
9     Case 3
10        Print "press 3"
11    Case 4
12        Print "Press 4"
13    Case 5
14        Print "press 5"
15    Case Else
16        Print "Invalid digit"
17 End Select
18 End
```

The execution output in the 'Untitled' window is:

```
Press any number b/w 0-5
enter number? 5
press 5
```

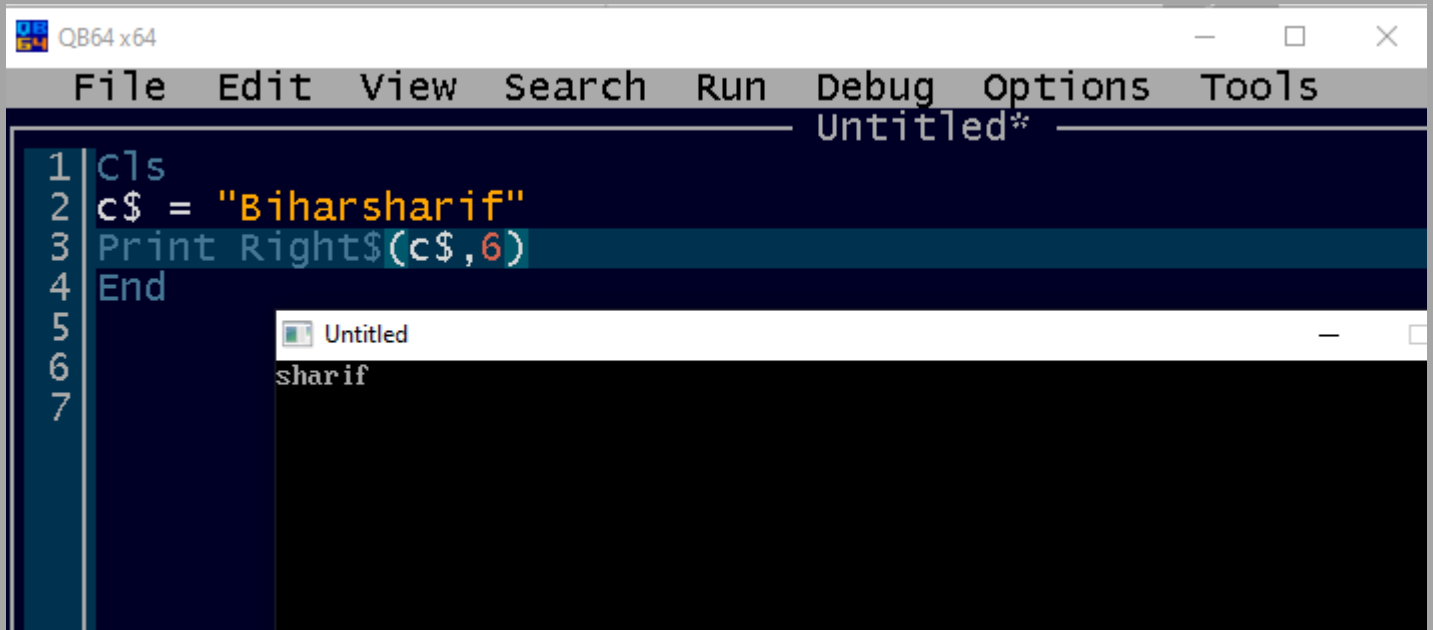
7. Return String

- This statement use return string character to Left side, Right side or Mid side.

Syntax

C\$ = "String"

Right/left/mid\$(c\$,digit) [Note digit – it is decide character return from where.]

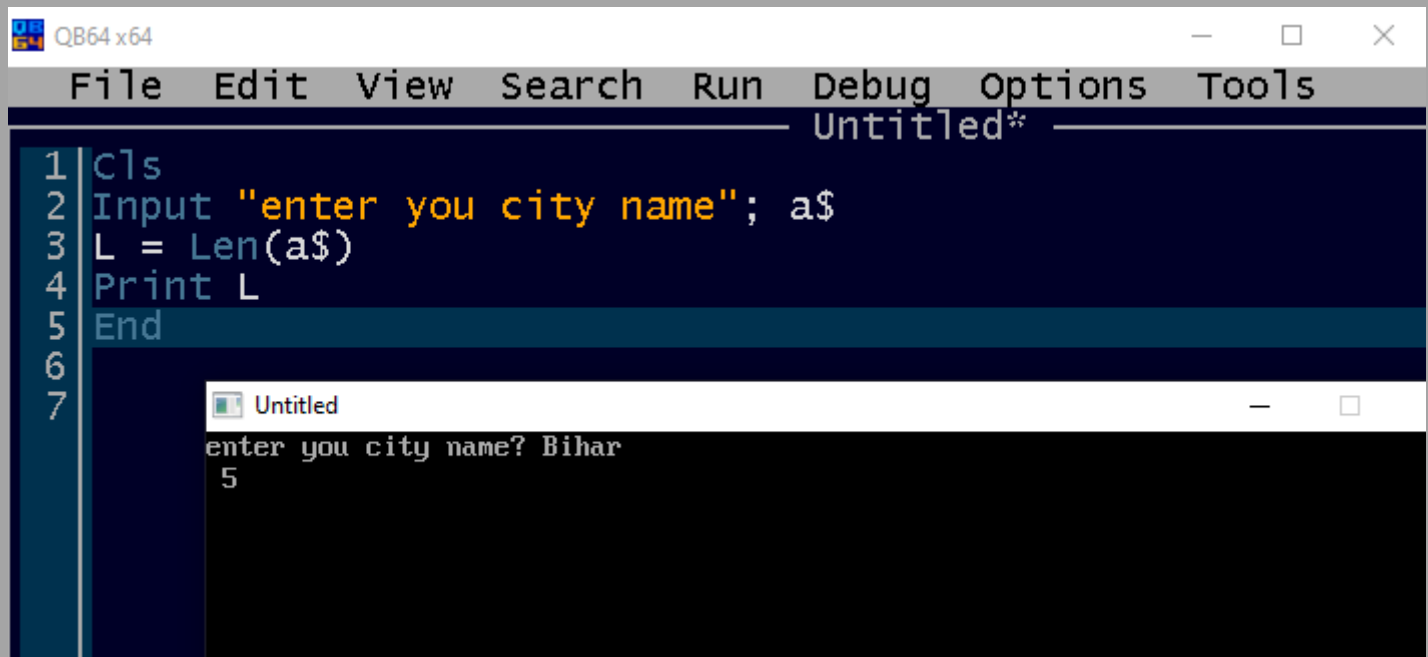


A screenshot of the QB64 x64 IDE. The main editor window, titled 'Untitled*', contains the following code:

```
1 Cls
2 c$ = "Biharsharif"
3 Print Right$(c$,6)
4 End
```

Below the editor, a console window titled 'Untitled' displays the output of the program: 'sharif'.

Print Number of Characters

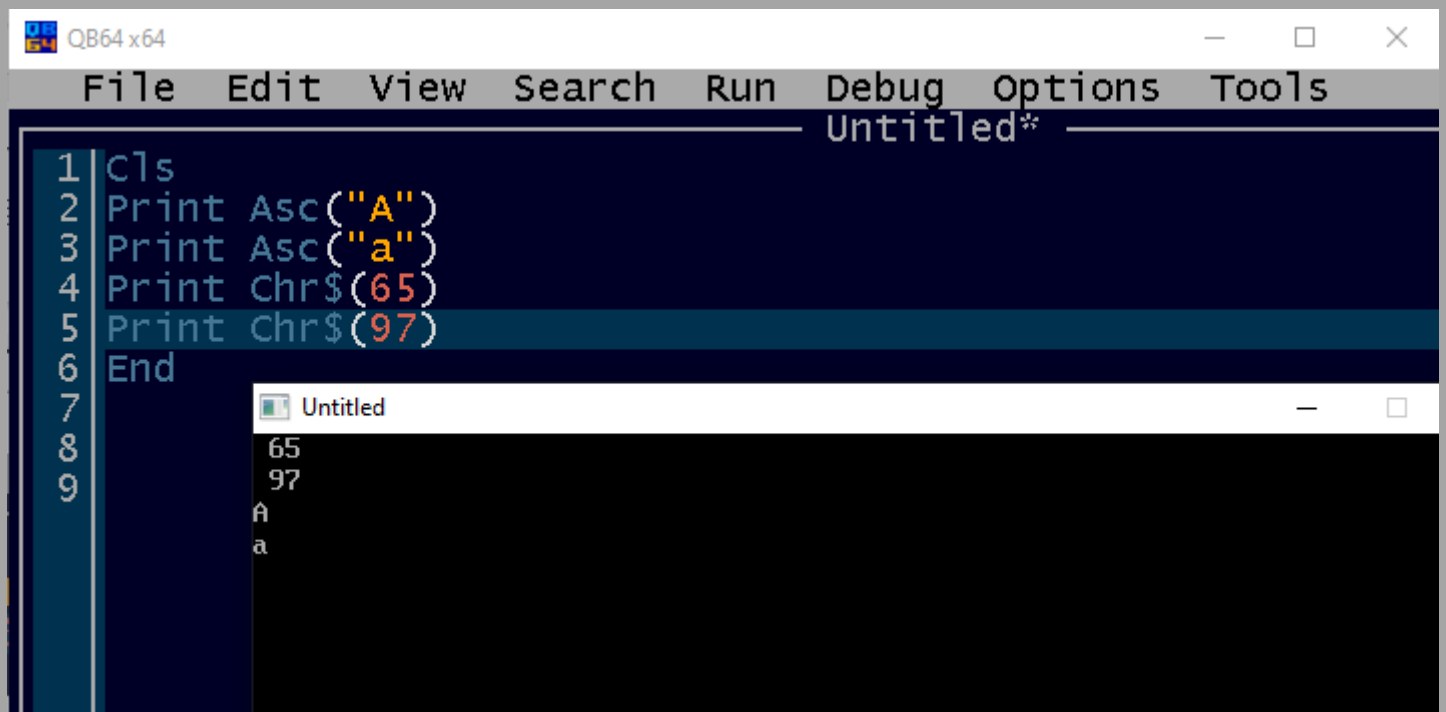


A screenshot of the QB64 x64 IDE. The main editor window, titled 'Untitled*', contains the following code:

```
1 Cls
2 Input "enter you city name"; a$
3 L = Len(a$)
4 Print L
5 End
```

Below the editor, a console window titled 'Untitled' displays the output of the program: 'enter you city name? Bihar' followed by '5'.

Print ASCII Value & Vice – versa



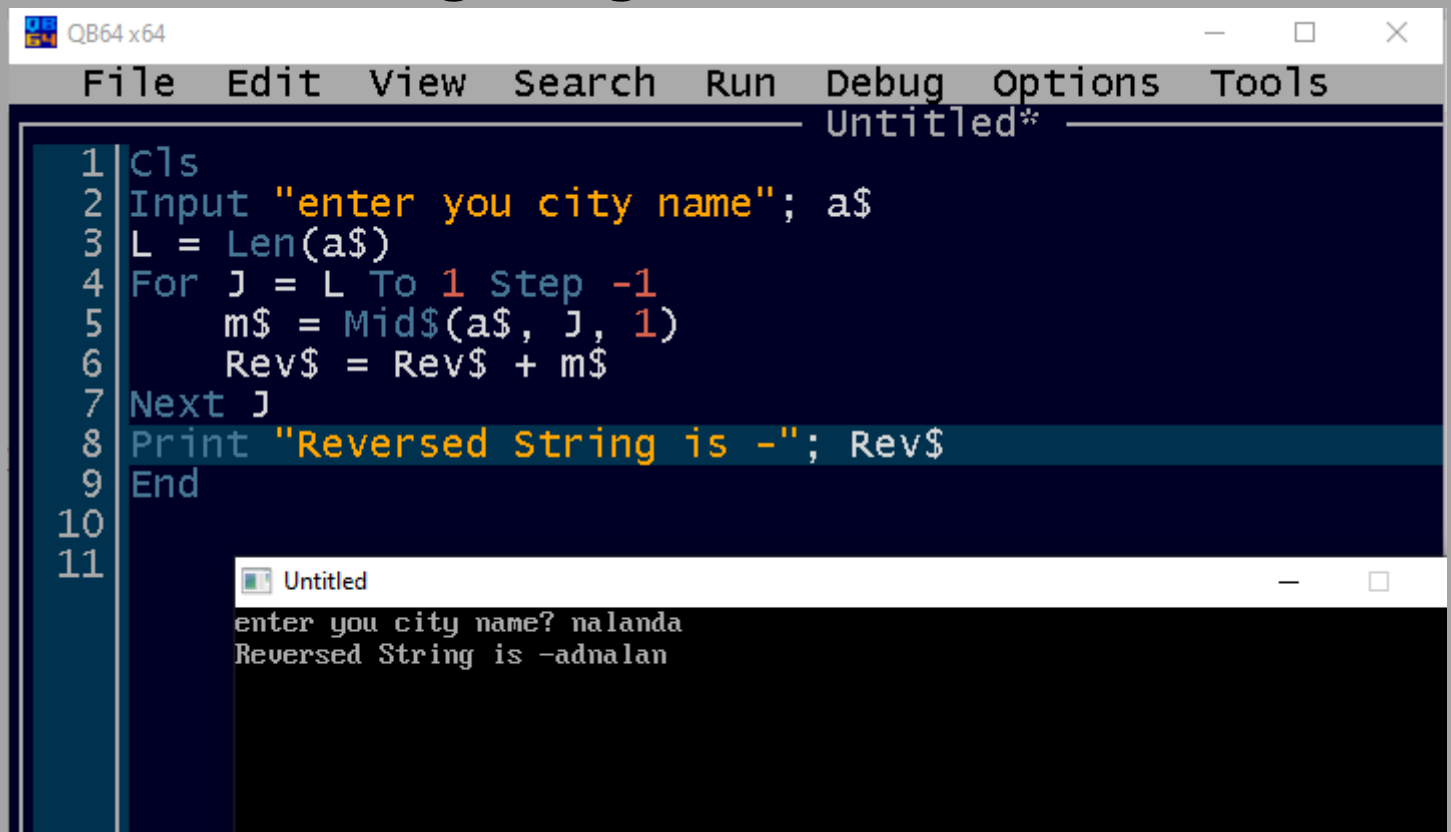
The screenshot shows the QB64 x64 IDE with a menu bar (File, Edit, View, Search, Run, Debug, Options, Tools) and a toolbar. The main editor window, titled 'Untitled*', contains the following code:

```
1 Cls
2 Print Asc("A")
3 Print Asc("a")
4 Print Chr$(65)
5 Print Chr$(97)
6 End
```

Below the main editor, a smaller window titled 'Untitled' displays the output of the program:

```
65
97
A
a
```

Reverse String Program



The screenshot shows the QB64 x64 IDE with a menu bar (File, Edit, View, Search, Run, Debug, Options, Tools) and a toolbar. The main editor window, titled 'Untitled*', contains the following code:

```
1 Cls
2 Input "enter you city name"; a$
3 L = Len(a$)
4 For J = L To 1 Step -1
5     m$ = Mid$(a$, J, 1)
6     Rev$ = Rev$ + m$
7 Next J
8 Print "Reversed String is -"; Rev$
9 End
```

Below the main editor, a smaller window titled 'Untitled' displays the output of the program:

```
enter you city name? nalanda
Reversed String is -adnalan
```